Machine Learning MCA 5121

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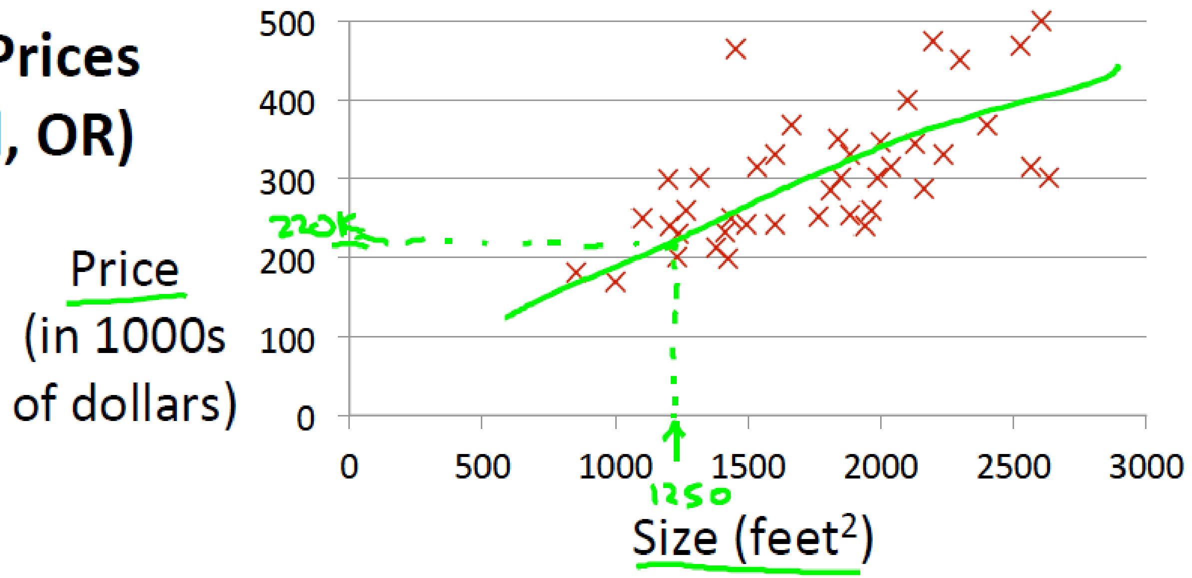
Slide - Set 2 – Regression Models

Supervised Learning
Linear Regression - Prediction

Housing Prices (Portland, OR)



<u>Supervised Learning</u>

Given the "right answer" for each example in the data.

<u>Regression Problem</u>

Predict real-valued output

Classification: Discrete-valued output

Andrew Ng

Vector Notations – Andrew NG

Training set of housing prices (Portland, OR)

Size in feet^2 (x)	Price ($) in 1000's (y)
2104	460
1416	232
1534	315
852	178
…	…

m = 47

Notation:

- **m** = Number of training examples
- **x**'s = "input" variable / features
- **y**'s = "output" variable / "target" variable

(x, y) – one training example

$(x^{(i)}, y^{(i)})$ – i^{th} training example

$x^{(1)} = 2104$

$x^{(2)} = 1416$

$y^{(1)} = 460$

Andrew Ng

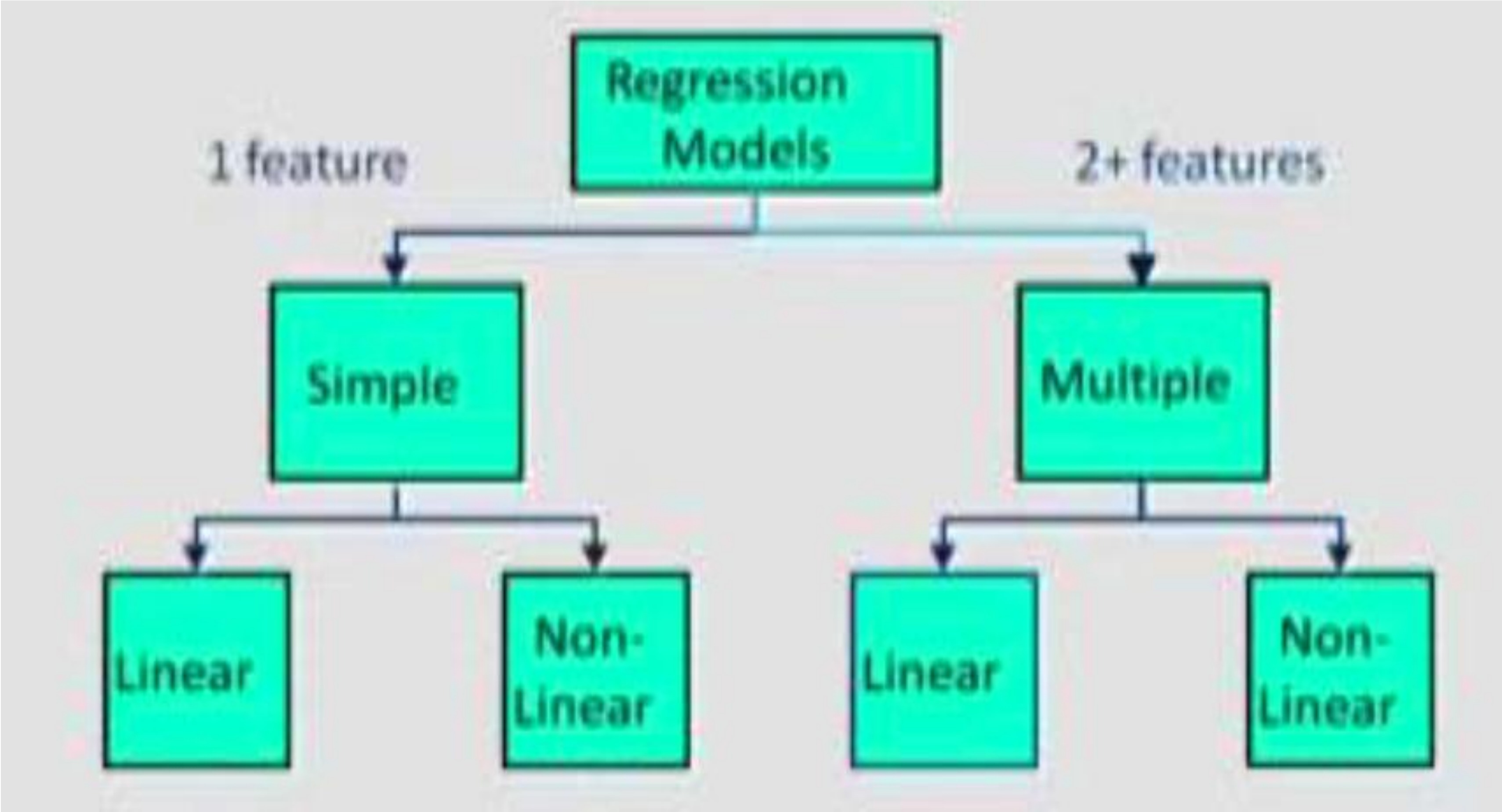
Regression Terminology

- Feature Space represents the data in a set of n features
- Features are properties that describe each instance in Instance space
- Multiple features represented in a Feature Vector
- We are given data and induction identifies a function, which can explain the data.
- Hypothesis could be a function which is a line or curve which separates classes.
- Hypothesis Space
  - is the set of all legal functions that are solutions to the task
  - Comprises of the features chosen and the language or the class of functions

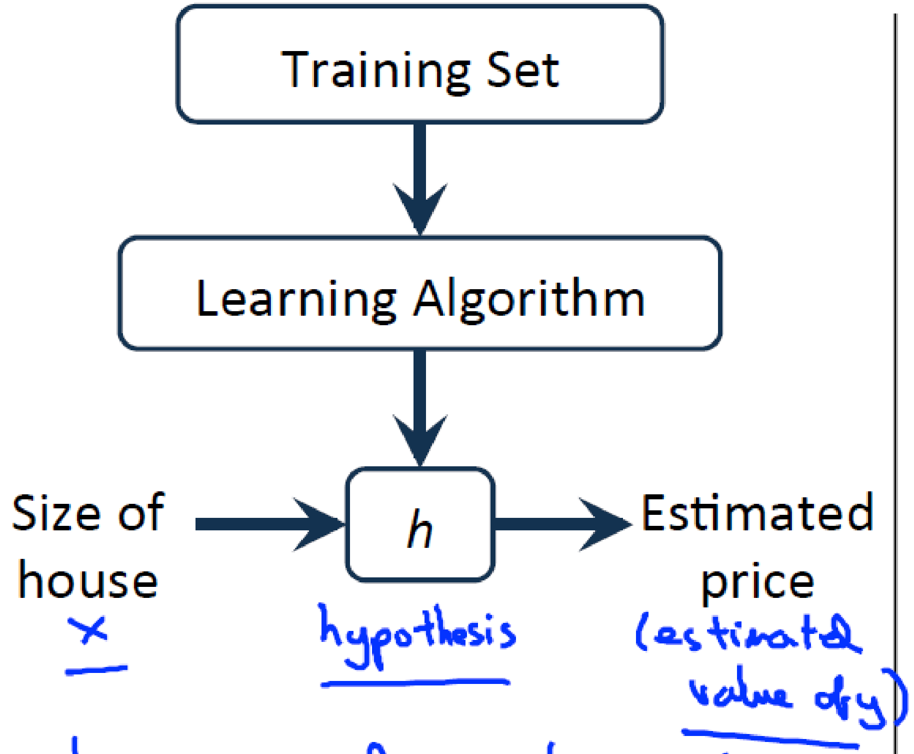
Regression Analysis

- Parametric Model
- Is a form of predictive modelling technique which investigates the relationship between
  - a **dependent** (target)
  - **independent variable (s)** (predictor).
- Used for forecasting, time series modelling and finding the causal effect relationship between the variables
- fit a curve / line to the data points, so that the differences between the distances of data points from the curve or line is minimized.

Type of Regression Models



Regression Hypothesis

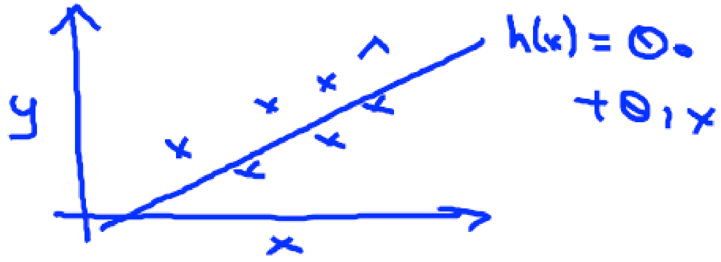


h maps from x's to y's.

How do we represent *h* ?

$h_\theta(x) = \underline{\theta_0 + \theta_1 x}$

Shorthand: $h(x)$



Linear regression with one variable. (x)

Univariate linear regression.

└ one variable

Regression Hypothesis

Training Set

Size in feet^2 (x)	Price ($) in 1000's (y)
2104	460
1416	232
1534	315
852	178
...	...

$m = 47$

Hypothesis: $h_\theta(x) = \theta_0 + \theta_1 x$

θ_i's: Parameters

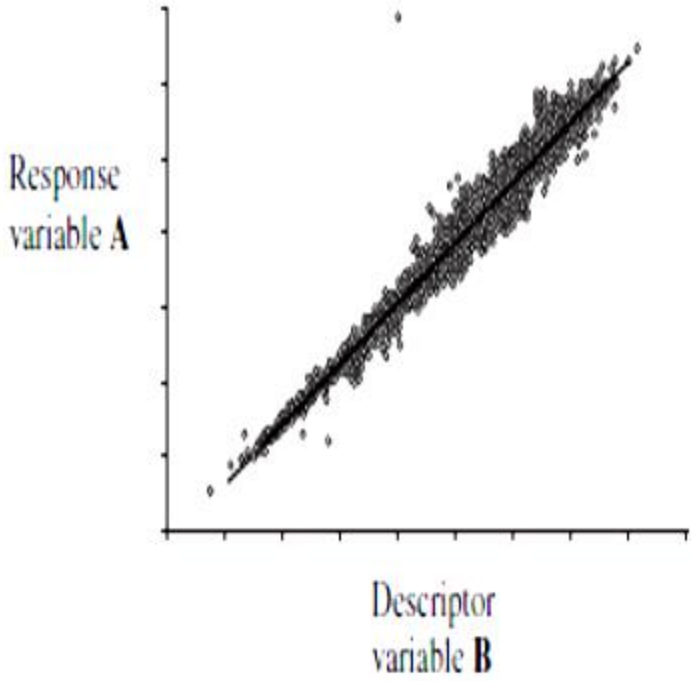
How to choose θ_i's ?

Simple Linear Regression - MLS

- Mathematical model that predicts continuous response variable
- Where there appears to be a linear relationship between two variables
- The prediction model is an equation of a straight line
- Method of Least squares

$$h_\theta(x) = \Theta_0 + \Theta_1 x$$

- $\Theta_1 = \dfrac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$
- $\Theta_0 = \bar{y} - \Theta_1 \bar{x}$



Comparative Analysis
Correlation Coefficient for numeric variables

- Generates values ranging from -1.0 to +1.0
- The value of r reflects how close to straight line the points lie.
- Positive numbers indicate a positive correlation
- Negative numbers indicate a negative correlation.
- If r is around 0 then there appears to be little or no relationship between the variables

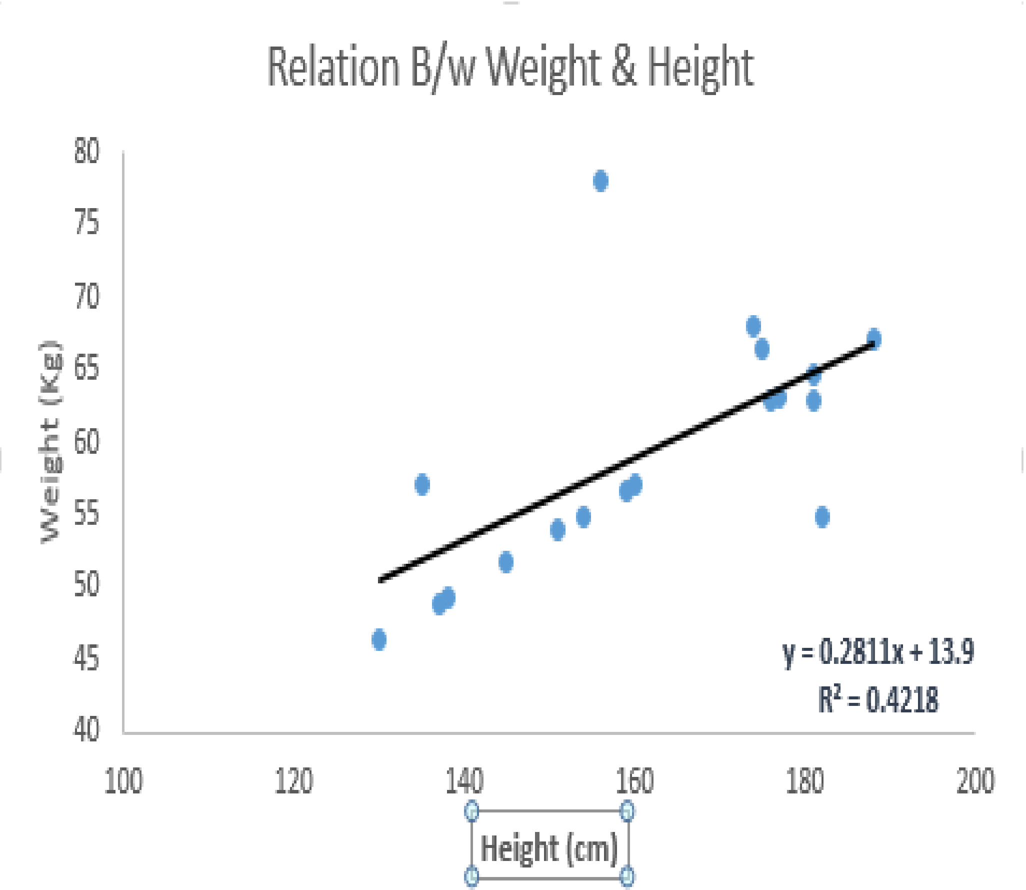
$$r = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{(n-1)s_x s_y}$$

Numerical 1 – Linear Regression - MLS

$X^{(i)}$	$Y^{(i)}$
2	69
9	98
5	82
5	77
3	71
7	84
1	55
8	94
6	84
2	64

1. Plot the values in a scatter plot
2. Compute Pearson's Correlation Coefficient
3. Compute a Simple Linear ML model using the method of least squares
4. Testing
   - If x=4, predict y?
   - If x=9, predict y?

Linear Regression

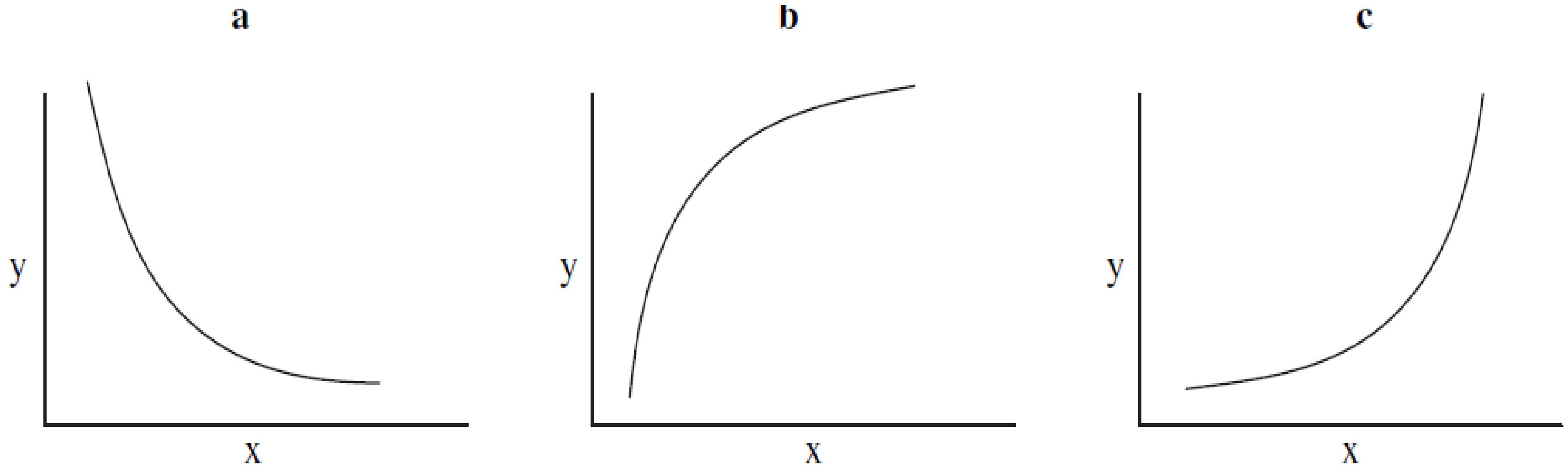


- There must be linear relationship between independent and dependent variables
- Relationship as a best fit line

  $y = \theta_0 + \theta_1 x + \varepsilon$
- we can assume that there is some noise in the data which causes Random error Ɛ
- Error is normally distributed with mean 0, and standard deviations σ
- Called as Gaussian noise or white noise
- To get best fit line use **LEAST SQUARE METHOD**
  - It calculates the best-fit line for the observed data by minimizing the sum of the squares of the vertical deviations from each data point to the line.
- Linear Regression is very sensitive to Outliers.
- **Assumptions**
  - there is very little or no multi-collinearity in the data.
  - There is very little or no auto-correlation in the error terms.
  - The error terms must possess constant variance.

Simple non-linear regression

- transform the nonlinear relationship to a linear relationship using a mathematical transformation
  - Situation a: Transformations on the x, y or both x and y variables such as log or square root.
  - Situation b: Transformation on the x variable such as square root, log or -1/x.
  - Situation c: Transformation on the y variable such as square root, log or -1/y.



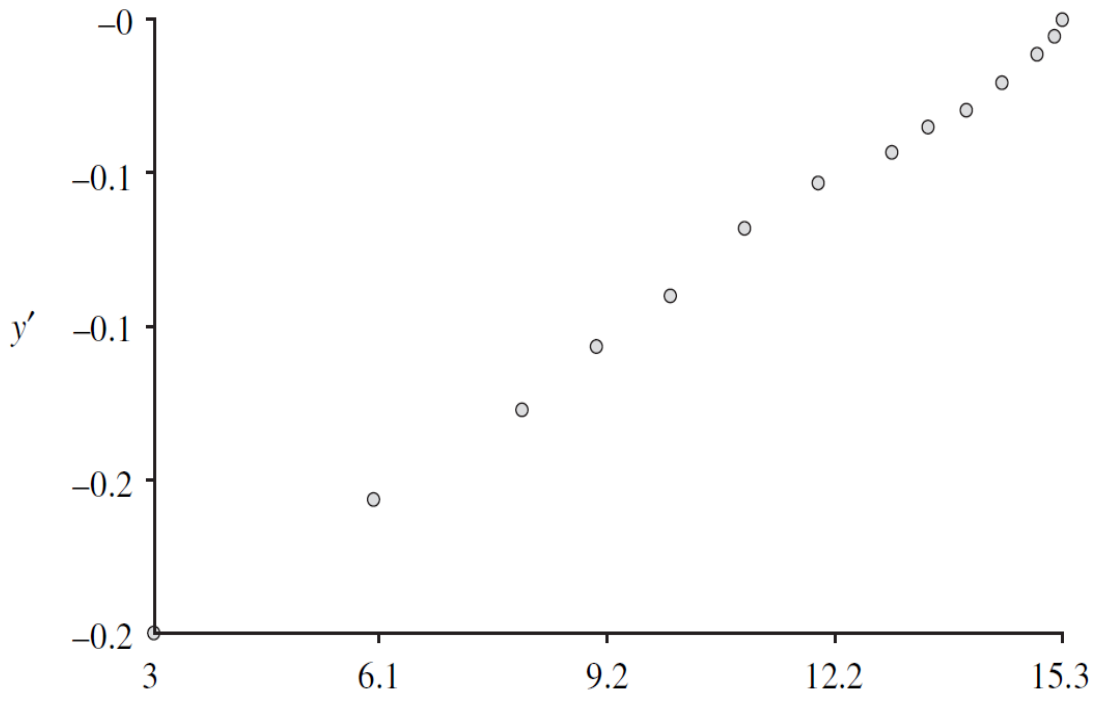
Numerical 2 – Simple Non-Linear Regression

x	y
3	4
6	5
9	7
8	6
10	8
11	10
12	12
13	14
13.5	16
14	18
14.5	22
15	28
15.2	35
15.3	42

1. Plot the values in a scatter plot
2. Compute a Simple Non Linear ML model using an appropriate transformation

Numerical 2 – Simple Non-Linear Regression Transformation

X	y	$y' = -1/y$
3	4	−0.25
6	5	−0.2
9	7	−0.14286
8	6	−0.16667
10	8	−0.125
11	10	−0.1
12	12	−0.08333
13	14	−0.07143
13.5	16	−0.0625
14	18	−0.05556
14.5	22	−0.04545
15	28	−0.03571
15.2	35	−0.02857
15.3	42	−0.02381



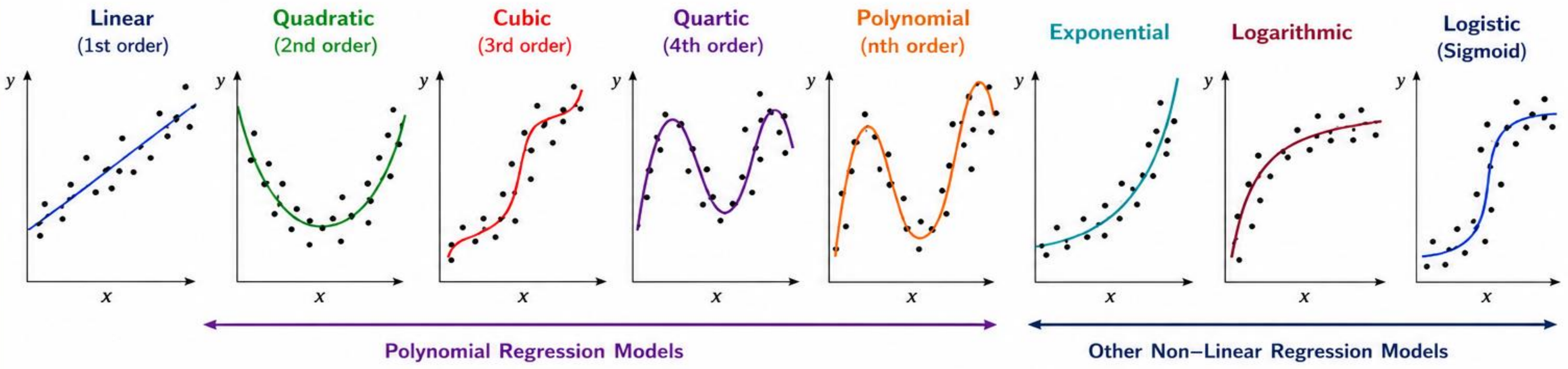
Numerical 2 – Simple Non-Linear Regression

$$y' = -0.307 + 0.018 \times x$$

x	y	$y' = -1/y$
3	4	−0.25
6	5	−0.2
9	7	−0.143
8	6	−0.167
10	8	−0.125
11	10	−0.1
12	12	−0.083
13	14	−0.071
13.5	16	−0.062
14	18	−0.056
14.5	22	−0.045
15	28	−0.036
15.2	35	−0.029
15.3	42	−0.024

If x = 9, what is y?
If x= 13.5 what is y?

Examples of Non–Linear Regression Models



Model Equations (Using θ Notation)

Model	Equation (using θ)	Description	Model	Equation (using θ)	Description
Linear	$y = \theta_0 + \theta_1 x + \varepsilon$	Straight line	Exponential	$y = \theta_0 e^{\theta_1 x} + \varepsilon$	Rapid growth or decay
Quadratic	$y = \theta_0 + \theta_1 x + \theta_2 x^2 + \varepsilon$	Parabolic (U or ∩ shape)	Logarithmic	$y = \theta_0 + \theta_1 \ln(x) + \varepsilon$ $(x > 0)$	Quick increase then levels off
Cubic	$y = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \varepsilon$	S–shaped curve	Power	$y = \theta_0 x^{\theta_1} + \varepsilon$ $(x > 0)$	Curve through origin (if $\theta_0 = 0$)
Quartic (4^{th} order)	$y = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4 + \varepsilon$	Wavy curve with up to 3 turning points	Logistic (Sigmoid)	$y = \dfrac{\theta_0}{1 + e^{-\theta_1 (x - \theta_2)}} + \varepsilon$	S–shaped curve with upper & lower limits
Polynomial (n^{th} order)	$y = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \cdots + \theta_n x^n + \varepsilon$	Flexible curve; degree n controls complexity			

y = Dependent variable (response) x = Independent variable (predictor) $\theta_0, \theta_1, \theta_2, \ldots, \theta_n$ = Model parameters ε = Error term

Experimental Model Evaluation for Prediction

- How is error measured? Suppose we want to make a prediction of a value for a target feature on example x
  - y is the observed value of the target feature on example x
  - ŷ is the predicted value of the target feature on example x
  - ŷ = h(x)
  - For Regression Problems
  - Mean Absolute Error = $\frac{1}{n}\sum_{i=1}^{n} |h(x) - y|$
  - Mean Square Error = $\frac{1}{n}\sum_{i=1}^{n}(h(x) - y)^2$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(h(x^{(i)}) - y^{(i)}\right)^2}$$

Numerical 3- Linear Regression – Error Computation using test data

$X^{(i)}$	Actual $Y^{(i)}$
3	75
4	72
7	87
8	95

1. Use the Regression Equation derived in Numerical 1 and compute the following:
2. Predicted Y for each X.
3. Computer Error
   a. Mean Absolute Error
   b. Mean Squared Error
   c. RMSE
   d. R^2

Optimising Weights with Cost Function

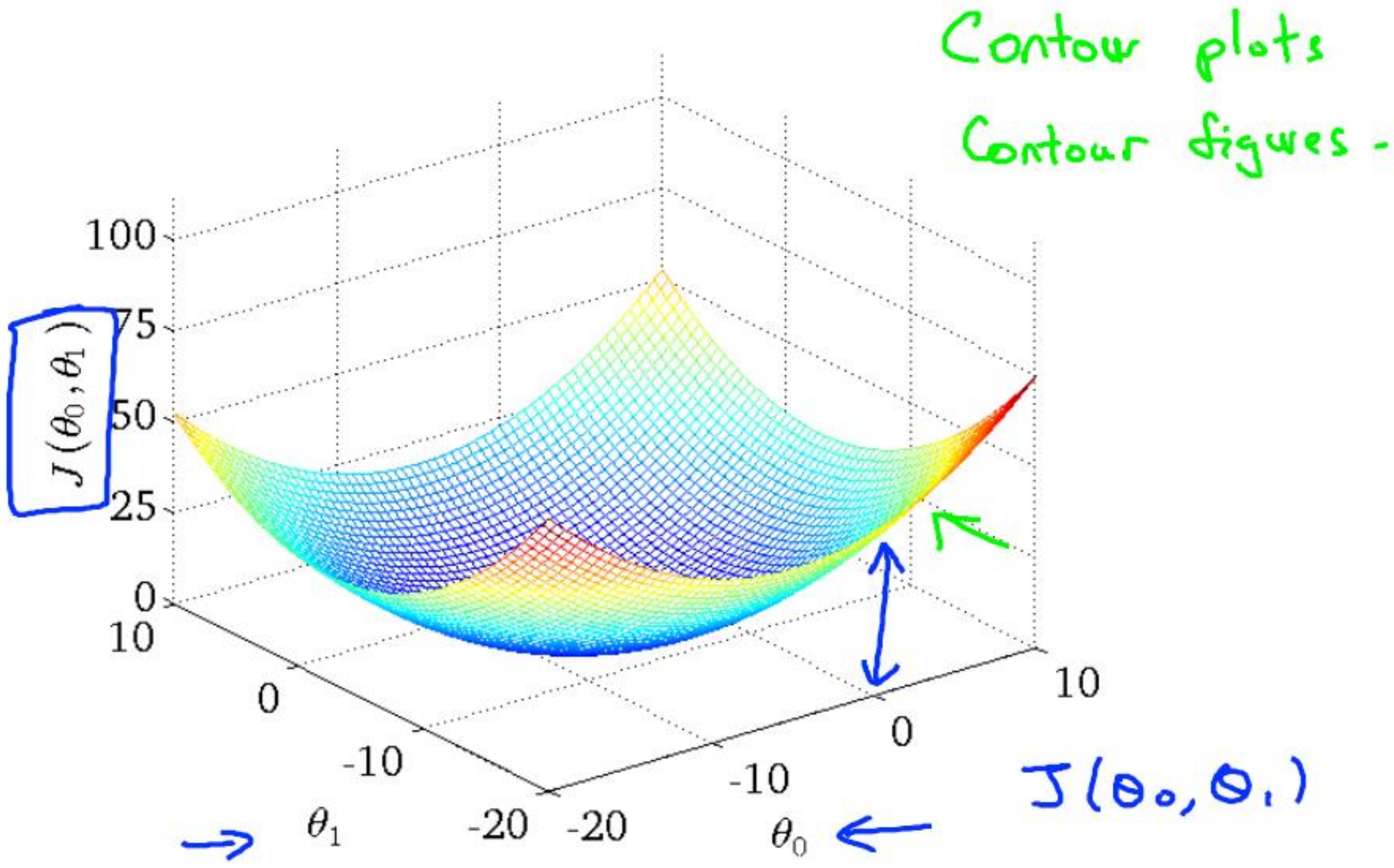
Hypothesis: $h_\theta(x) = \theta_0 + \theta_1 x$

Parameters: θ_0, θ_1

Cost Function: $J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)}\right)^2$

Goal: $\underset{\theta_0, \theta_1}{\text{minimize}} \; J(\theta_0, \theta_1)$

3D Surface plot



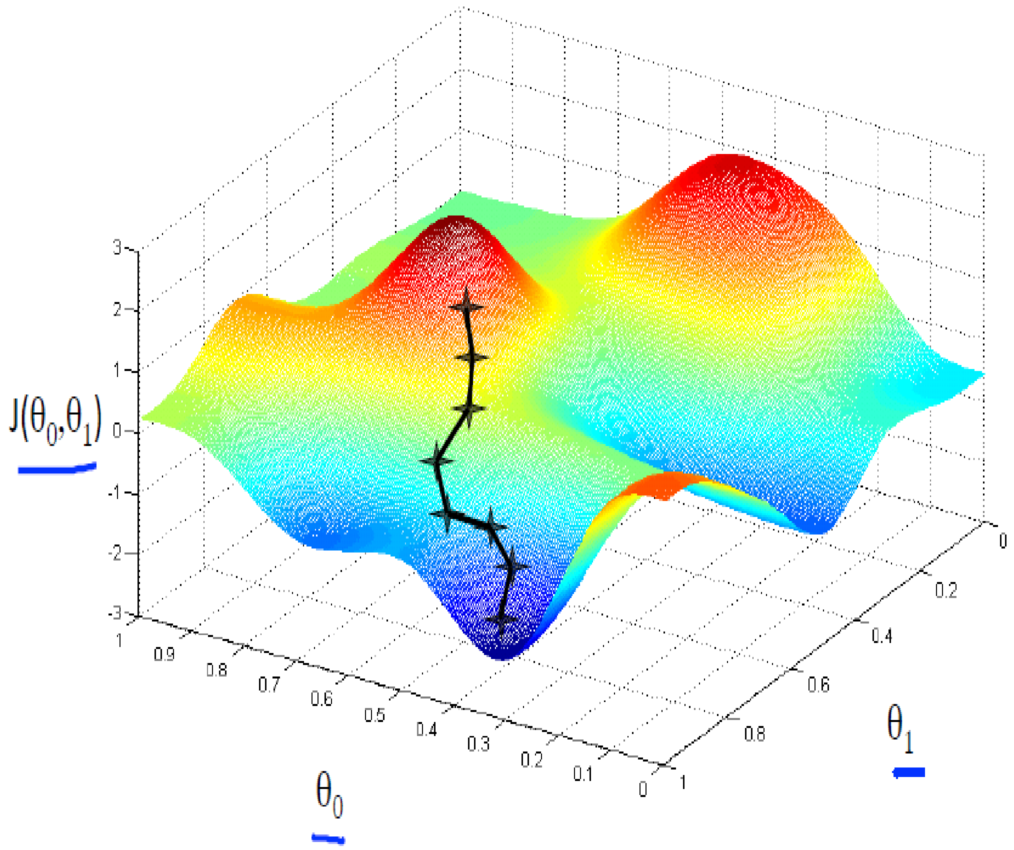
Gradient Descent

Have some function $J(\theta_0, \theta_1)$ $J(\theta_0, \theta_1, \theta_2, \dots, \theta_n)$

Want $\min_{\theta_0, \theta_1} J(\theta_0, \theta_1)$ $\min_{\theta_0, \dots, \theta_n} J(\theta_0, \dots, \theta_n)$

Outline:

- Start with some θ_0, θ_1 (say $\theta_0 = 0, \theta_1 = 0$)
- Keep changing θ_0, θ_1 to reduce $J(\theta_0, \theta_1)$ until we hopefully end up at a minimum



Gradient descent algorithm

$$\text{repeat until convergence } \{$$

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)$$

$$(\text{for } j = 1 \text{ and } j = 0)$$

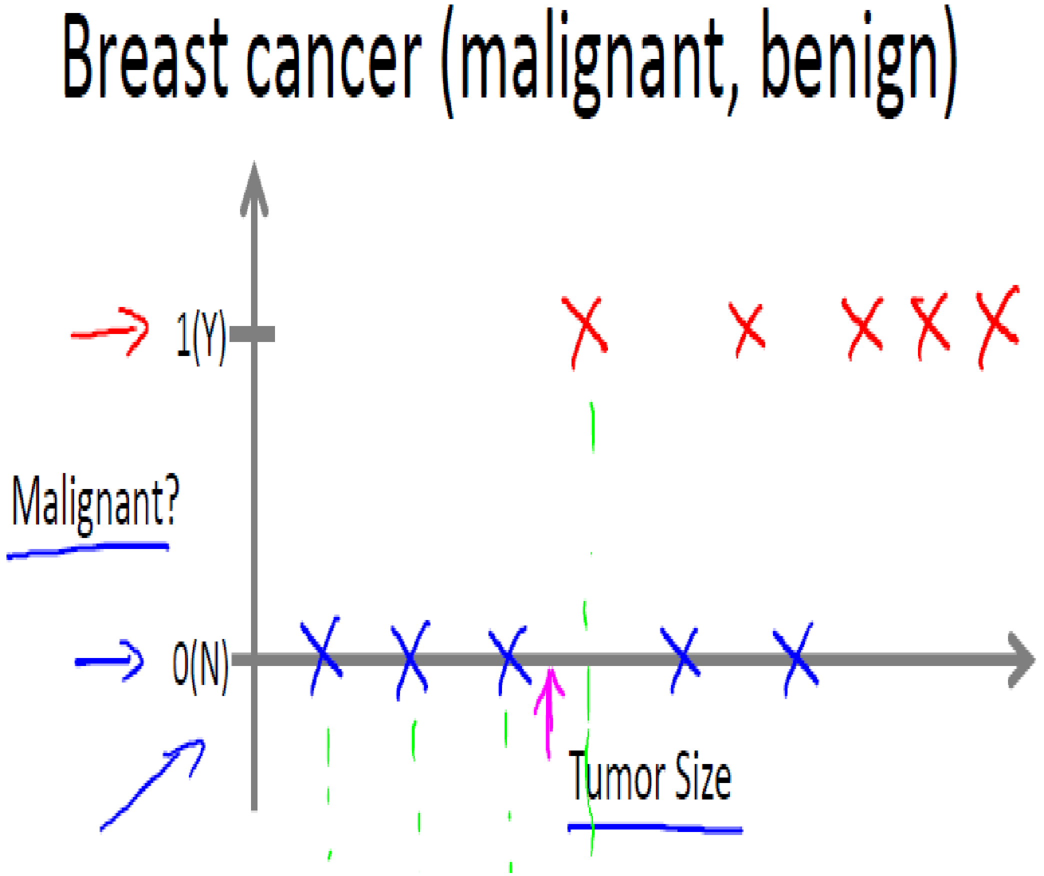
$$\}$$

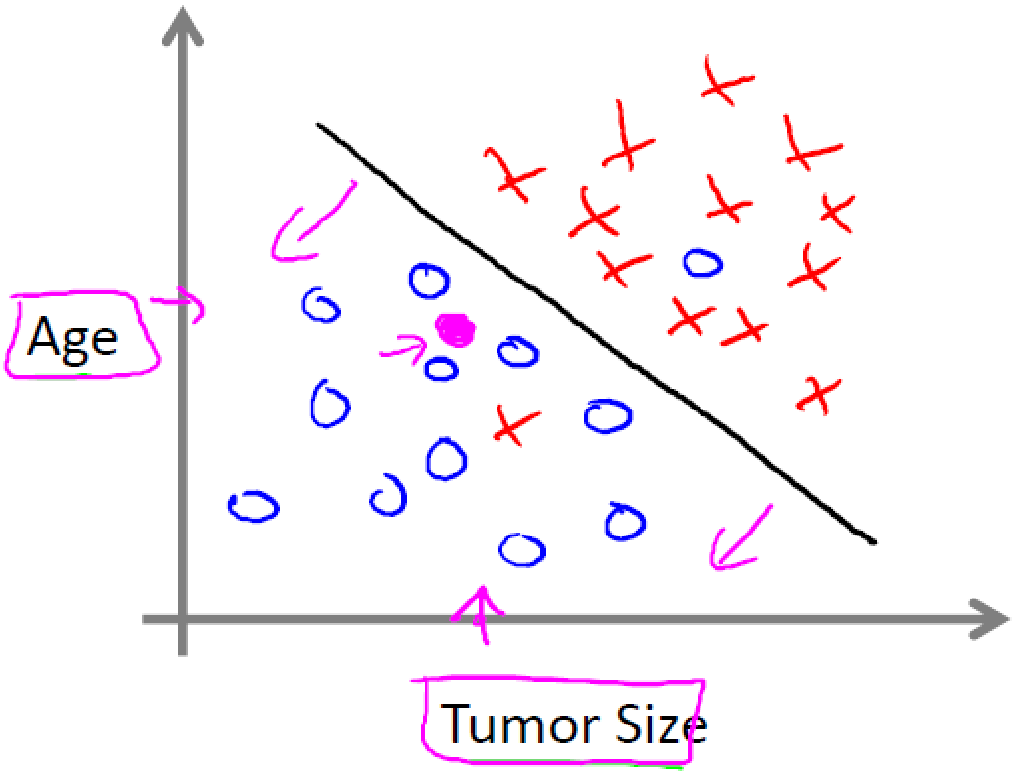
Linear Regression Model

$$h_\theta(x) = \theta_0 + \theta_1 x$$

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)}\right)^2$$

Supervised Learning
Logisitic Regression - Classification

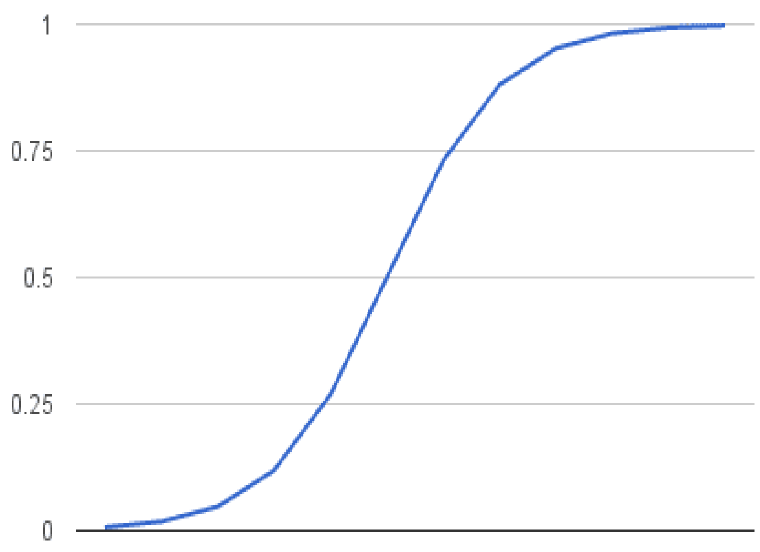




Numerical 4- Logistic Function

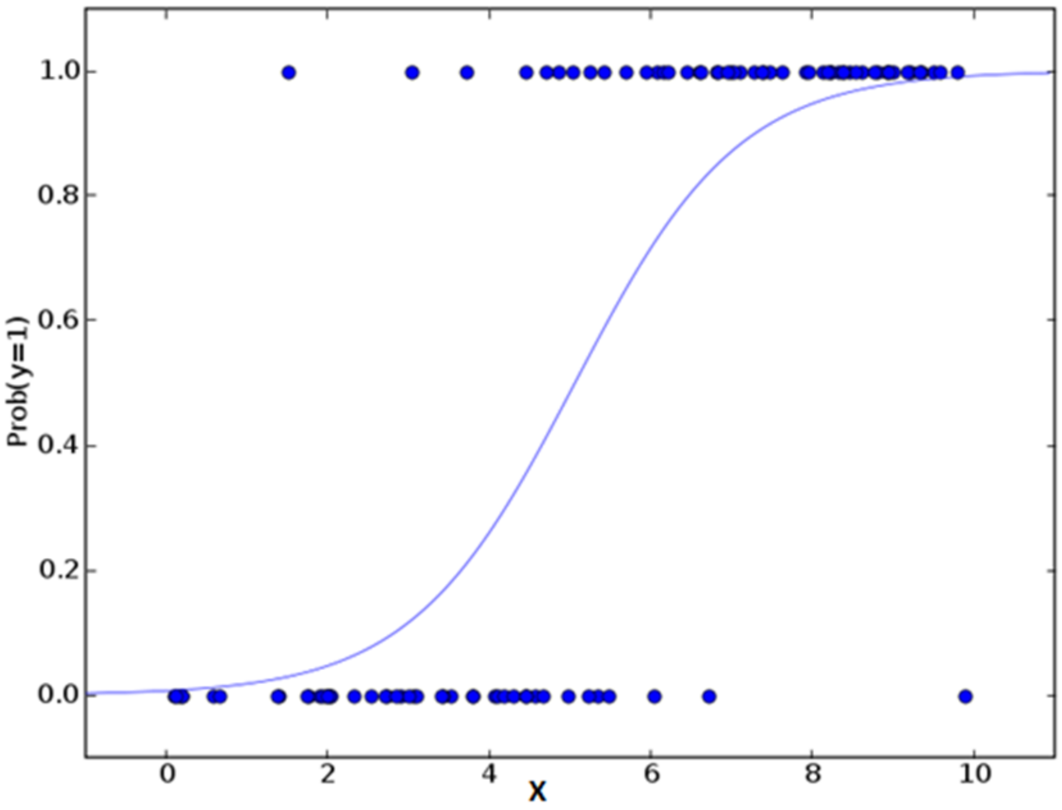
- The logistic function is defined as: transformed = 1 / (1 + e^-x)

X	Transformed y
-5	
-4	
-3	
-2	
-1	
0	
1	
2	
3	
4	
5	



Logistic Regression for classification

- Classification
  - Email – Spam/Not Spam
  - Tumor –is Malignant/Benign
- y Ɛ {1,0} - 1 is positive class and 0 is negative class
- Can be extended to y Ɛ {0,1,2,3}
- Threshold classifier output $h_\Theta(x)$ at 0.5
  - If $h_\Theta(x) >= 0.5$ then y = 1
  - If $h_\Theta(x) < 0.5$ then y = 0
- Logistic Regression :
  - $0 <= h_\Theta(x) <= 1$



Logistic Regression – hypothesis function

Hypothesis Function (Sigmoid or Logistic Function)

$$h_\theta(x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}$$

- e is Euler's number, the base of the natural logarithm.
- θ_0 is the y-intercept (intercept term).
- θ_1 is the coefficient (weight) associated with input x.
- $h_\theta(x)$ represents the estimated probability that $y = 1$ for input x.

Interpretation:

$$h_\theta(x) = P(y = 1 \,|\, x; \theta)$$

$$P(y = 0 \,|\, x; \theta) = 1 - h_\theta(x)$$

Logistic Regression – Numerical 5

- Consider the following data set which predicts whether a customer is a ‘Loan Non-Defaulter’ based on the ‘Amount of Savings (in Lakhs)’. The logistic regression analysis on the training data set gave the following output.
  - The coefficients are θ_0= **-4.07778** and θ_1**= 1.5046.**
  - For the following test data set predict the class label using Logistic Regression
  - Compute the Confusion Matrix of the Classifier Model and comment on the Performance.

Savings (in Lakhs)	0.5	0.85	1	1.25	1.75	1.75	2.25	3.25	4.25	5
Non-Defaulter	0	0	0	0	0	1	1	1	1	1

Numerical 5 – Computing predicted class

$$h_\theta(x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x)}}$$

Savings (in Lakhs)	0.5	0.85	1	1.25	1.75	1.75	2.25	3.25	4.25	5
Non-Defaulter (Actual)	0	0	0	0	0	1	1	1	1	1
(Predicted)	0 (0.04)	0 (0.05)	0 (0.07)	0 (0.10)	0 (0.19)	0 (0.19)	0 (0.33)	1 (0.69)	1 (0.91)	1 (0.97)

Numerical 5- Evaluating accuracy

- Confusion Matrix

Actual vs Predicted	Predicted as 1	Predicted as 0	Row Total
Actual 1	(TP)	(FN)	
Actual 0	(FP)	(TN)	
Column Total			

Compute the metrics of

- Accuracy
- Sensitivity (TPR)
- Specificity (TNR)
- Precision
- Recall

Numerical 5- Evaluating accuracy

- **P = TP + FN** → Total Actual Positives
- **N = TN + FP** → Total Actual Negatives
- **P + N** → Total number of samples
- **Accuracy = (TP + TN) / (P + N)**
- **Sensitivity or Recall (TPR) = TP / P** [Measures how well the model finds all real +ve cases.]
- **Specificity (TNR) = TN / N**
- **Precision = TP / (TP + FP)** [Measures how reliable a positive prediction is.]
- **F1-Score = 2 *** $\frac{\boldsymbol{Precision * Recall}}{\boldsymbol{Precision + Recall}}$

Evaluating Model Performance Beyond Accuracy

Scenario: Credit Card Fraud Detection (1,000 Total Transactions | 950 Legitimate vs. 50 Fraudulent)

	Pred Normal	Pred Fraud	Total Actual
Actual Normal	920 (TN)	30 (FP)	950
Actual Fraud	10 (FN)	40 (TP)	50

Accuracy

96.0%

Formula: (40 + 920) / 1000

Misleading metric. Guessing 'Normal' every single time yields 95% accuracy while catching zero fraud.

Precision

57.1%

Formula: 40 / (40 + 30)

Reliability of alerts. Out of 70 flagged transactions, only about 57% are actual fraud.

Recall

80.0%

Formula: 40 / (40 + 10)

Coverage of actual threats. The system successfully caught 80% of all real fraud cases.

F1 Score

66.6%

Formula: Harmonic Mean

The balanced view. Realistically reflects poor precision holding back good overall recall.

Core Takeaway

Never rely on accuracy alone for imbalanced data. Use Precision to track false alarm costs, Recall for missed threats, and F1 Score for a true, balanced evaluation baseline.

Numerical 6- Model Comparison

- Compare the 3 models using metrics of Accuracy, TPR and TNR

Model A

Actual

Predicted	1	0	Totals
1	79	28	107
0	72	213	285
Totals	151	241	392

Model B

Actual

Predicted	1	0	Totals
1	140	38	178
0	11	203	214
Totals	151	241	392

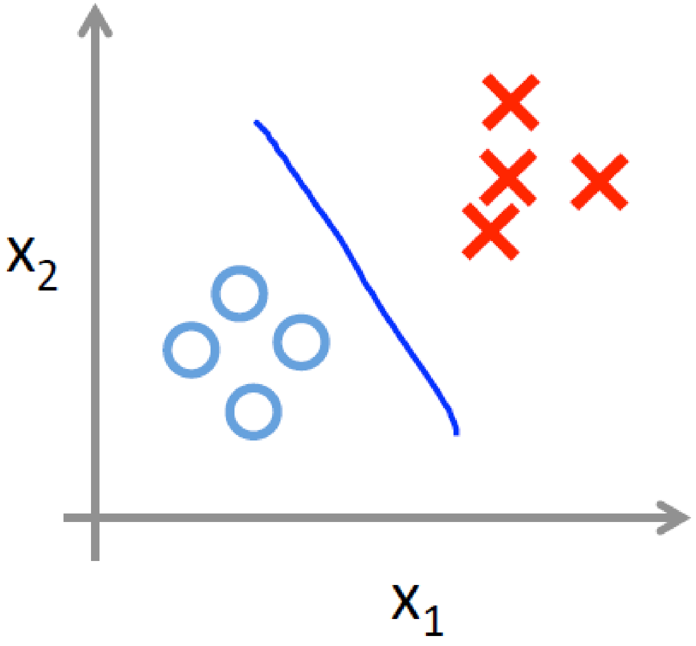
Model C

Actual

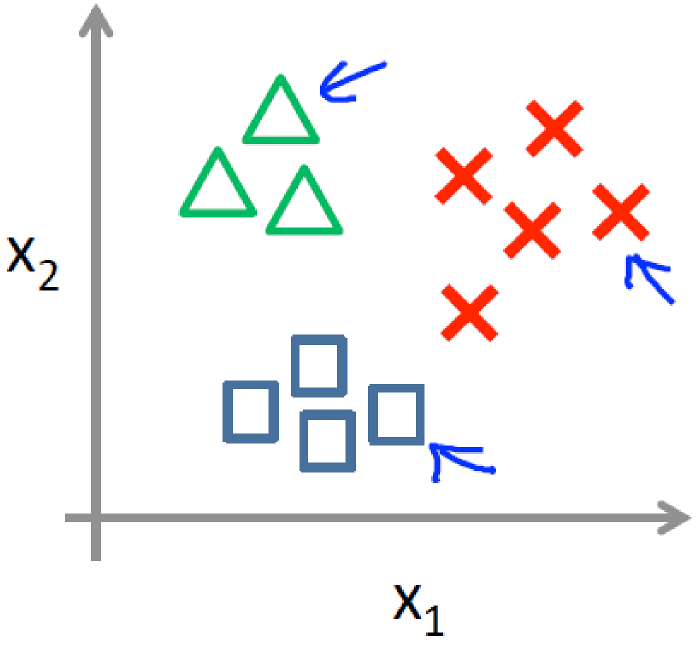
Predicted	1	0	Totals
1	129	18	147
0	22	223	245
Totals	151	241	392

Multiclass Classification

Binary classification:



Multi-class classification:



$$h_\theta^{(i)}(x) = P(y = i | x; \theta) \qquad (i = 1, 2, 3)$$

Multi class Classification

One-vs-all

Train a logistic regression classifier $\underline{h_\theta^{(i)}(x)}$ for each class $\underline{i}$ to predict the probability that $\underline{y = i}$.

On a new input $\underline{x}$, to make a prediction, pick the class i that maximizes

$$\max_{\underline{i}} \underline{h_\theta^{(i)}(x)}$$

Multiclass Classification- Confusion Matrix

Binary: 2 × 2

	0	1
0	TN	FP
1	FN	TP

Iris: 3 × 3

	Setosa	Versicolor	Virginica
Setosa	TP Setosa	Setosa → Versicolor	Setosa → Virginica
Versicolor	Versicolor → Setosa	TP Versicolor	Versicolor → Virginica
Virginica	Virginica → Setosa	Virginica → Versicolor	TP Virginica

What changes in multiclass?

- Binary is 2×2; Iris is 3×3.
- Diagonal cells = correct predictions.
- Off-diagonal cells = class confusions.

Class-wise report metrics

Treat each Iris class as positive vs. the other two.

$\text{Precision}_i = TP_i / (TP_i + FP_i)$

$\text{Recall}_i = TP_i / (TP_i + FN_i)$

$F1_i$ = harmonic mean of precision and recall

Support_i = actual samples in class i

Precision: “How many positive predictions were correct?”

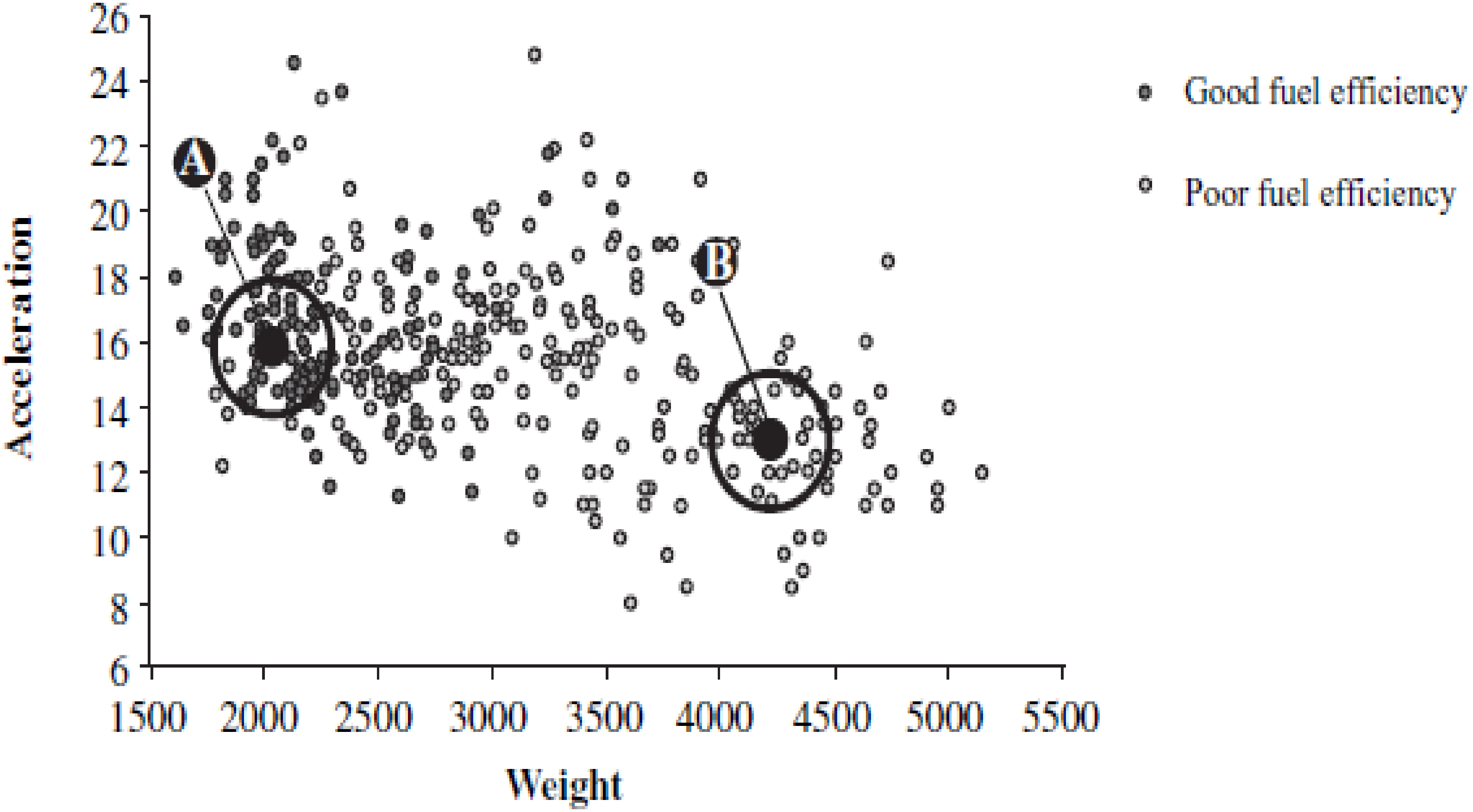
Recall: “How many actual positives did we find?”

F1=2×(Precision*Recall)/(Precision+Recall)

Eager Vs. Lazy Learners

- **Eager learners**
  - when given a set of training tuples, will construct a classification model before receiving test tuples to classify.
  - Example: Regression Models
- **Lazy learner**
  - simply stores it and waits until it is given a test tuple. Only when it sees the test tuple does it perform generalization to classify the tuple based on its similarity to the stored training tuples.
  - are also referred to as **instance-based learners**, even though all learning is essentially based on instances.
  - Can be computationally expensive , not suited for real time decisions
  - Lazy learners, however, naturally support incremental learning.
  - Example : k-nearest neighbor, Bayesian Classifiers

Visualization of k-nearest neighbors



K-Nearest Neighbors

- Training tuples are described by n attributes and each tuple represents a point in n dim space.
- Learning by analogy - By comparing a given test tuple with training tuples that are similar to it by defining its neighborhood in terms of “k”.
- Classifier searches the pattern space for the k training tuples that are closest to the unknown tuple.
- For k-nearest-neighbor classification , the unknown tuple is assigned the most common class among its k nearest neighbors.
- Distance measures could be Euclidean, manhattan or minkowski etc.
- Preprocessing may be required
  - If attributes have initial large ranges normalize the attributes.
  - Noisy data could affect results
- Advantages:
  - Large sets: kNN can be used with large training sets.
  - Instance based learning so no model is maintained.
- Disadvantage:
  - Since decision is made based on k neighbors instead of entire input space, susceptible to noise,if k is small

Distance Measures for numeric attributes

- Euclidean Distance
  - $X_1 = (x_{11},x_{12},\ldots x_{1n})$ and $X_2 = (x_{21},x_{22}..x_{2n})$ is
  - $dist(X_1,X_2) = \sqrt{}\Sigma_{1=1}^{n} (x_{1i} - x_{2i})^2$

- Manhattan or City Block Distance
  - $dist(i , j) = | x_{i1} - x_{j1}| + | x_{i2} - x_{j2}| +\ldots | x_{in} - x_{jn}|$

- Minkowski Distance
  - $dist(i,j) = (| x_{i1} - x_{j1}|^p + | x_{i2} - x_{j2}|^p+\ldots | x_{in} - x_{jn}|^p)^{1/p}$

Obs	Income in (1000s)	Plot Size	Owns LM	Distance	Rank
1	60	18.4	Y		
2	85.5	16.8	Y		
3	64.5	21.6	Y		
4	61.5	20.8	Y		
5	87	23.6	Y		
6	32	19.2	Y		
7	108	17.6	Y		
8	75	19.6	N		
9	52.8	20.8	N		
10	64.8	17.2	N		
11	43.2	20.4	N		
12	84	17.6	N		
13	49.2	17.6	N		
14	66	18.4	N		

Numerical 7-KNN Classification

A riding lawn mover manufacturer would like to classify families into those who are likely to purchase vs not likely.

For a Test Instance where
Income is 52
Plot size is 21

Predict if the family will purchase a lawn mower or not for k=1,2,3 using Manhattan distance
Use Majority Voting when necessary

Age	Loan (in 1000s)	HPI	Distance	Rank
25	40	135		
35	60	256		
45	80	231		
20	20	267		
35	120	139		
52	18	150		
23	95	127		
40	62	216		
60	100	139		
48	220	250		
33	150	264		

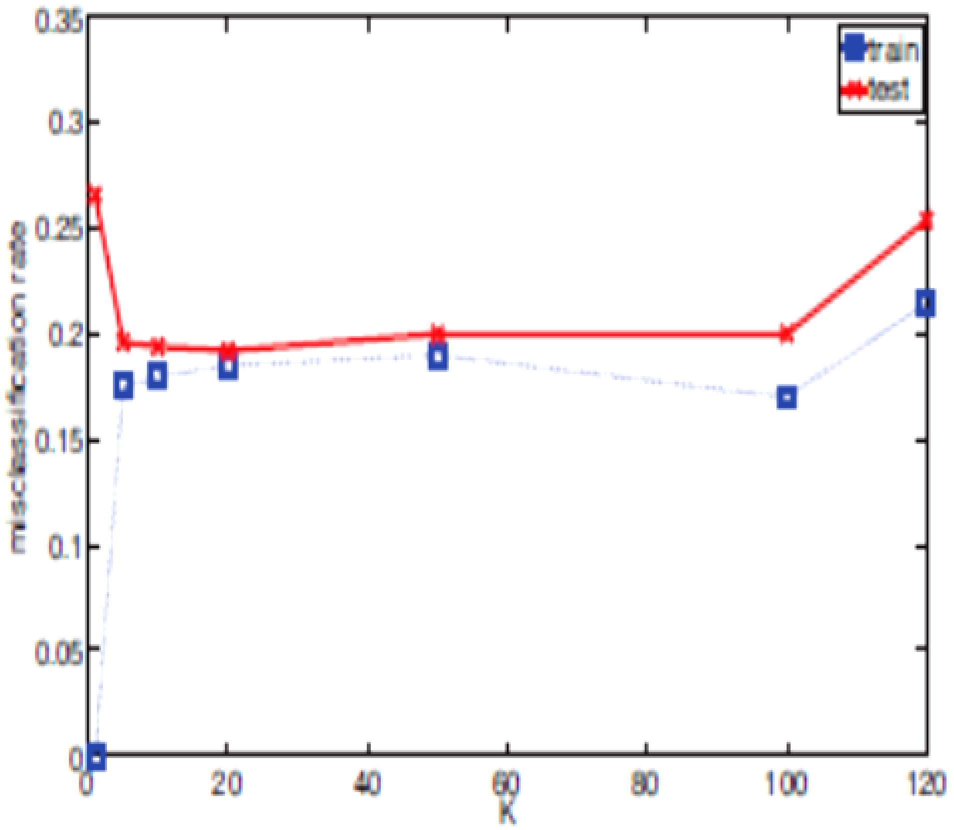
Numerical 8- KNN Prediction

Consider the following data concerning Housing Price Index

For a Test Instance where
Age = 48
Loan = 1,42
Predict the HPI using Euclidean Distance and
K=1,2,3
Use averaging method for Distance

Model Selection in the KNN case

- We care about generalization error or test error
- Test error (Red Line)
  - U-shaped curve
  - for complex models (small *K*), the method overfits
  - simple models (big *K*), the method underfits
- Use validation test to pick *K*
  - *Pick* the value with the minimum error on the test set
  - K = 10 to 100



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- Machine Learning by Andrew N G (Chapter 6)
  - https://www.youtube.com/watch?v=-la3q9d7AKQ